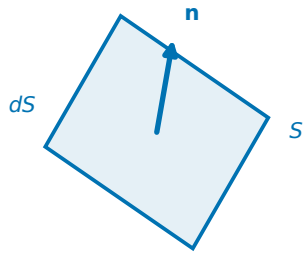
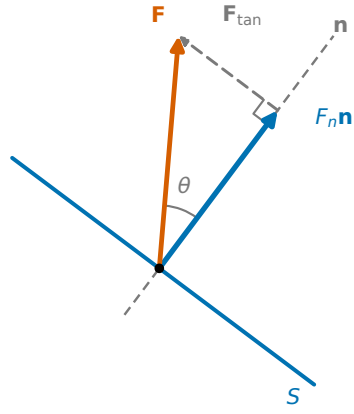
A



$$d\mathbf{S} = \mathbf{n}\, dS$$

B



$$F_n = \mathbf{F} \cdot \mathbf{n} = \|\mathbf{F}\| \cos\theta$$

$$\mathbf{F} \cdot d\mathbf{S} = (\mathbf{F} \cdot \mathbf{n})\, dS = F_n\, dS$$